

साक्षी

Saakshi

A resilient, provable computer-based testing ecosystem

Hackathon Challenge 6 · Solution submission

"We can't promise zero failures. We make every failure recoverable, contained, provable."

Executive summary

Saakshi ("witness") is trust infrastructure for India's high-stakes computer-based exams (JEE, NEET from 2027, CUET). Every answer is signed and hash-chained at the seat, end-to-end encrypted to the data centre, and countersigned on arrival, so no one in between can read, alter or lose it without leaving proof.

Monitor the machine

Integrity gate before T0;
kernel-level monitoring
(anti-cheat style) to catch
VMs and relay software.

Protect every answer

Signer + reader signatures
on a git-like hash chain;
end-to-end encryption the
centre cannot open.

Survive every failure

Relays buffer through
crashes and outages; a
cluster of 3 data-centre
cells rebuilds itself.

Decide fairly

Evidence locates
tampering and harm, so
only the affected
candidates are re-tested.

The solution at a glance

0 / 16,000

answers lost across 20 chaos runs

~1 s

median recovery time after killing or wiping a cell

1,000 / 1,000

one-byte tamper edits located to their entry

19,803

candidates unlocked in a 20k load run (of 19,803)

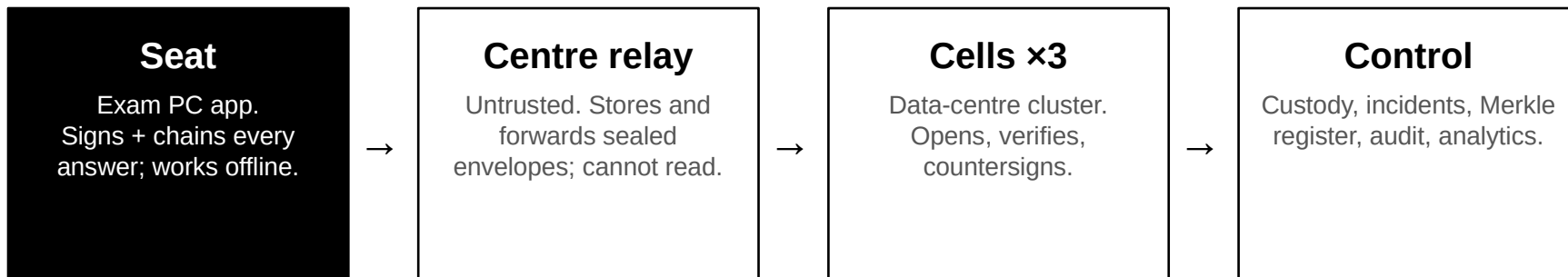
Who: NTA (policy, keys, decisions), NIC (data centres), exam vendors (centres), independent witnesses.

What: Seat app, untrusted centre relay, 3-cell data-centre cluster, control, witness; open protocol v1.

Status: Working prototype, tested on macOS and Windows CI; all 12 challenge focus areas traced to code and tests.

Ask: A shadow-mode pilot on one small NTA exam, then a conformance kit written into the CBT vendor RFP.

Concept: five parts, each distrusting the next



- Answers get WhatsApp-style ticks: ✓ saved on seat, ✓ ✓ relay has it, blue ✓ ✓ data centre countersigned.
- Question papers sit encrypted on disk; any 2 of 3 custodians (NTA, NIC, observer) release the key.
- A failure anywhere is contained to its own part and recovered from the parts around it.

Approach: the path of one answer

- 1 Candidate clicks B on Q17** Seat signs the entry header, links it to the previous entry's hash, encrypts the body to the cell, fsyncs. ✓
- 2 Relay receives** Checks signature and chain, stores durably, acks. It never sees "B". ✓✓
- 3 Cell reads and countersigns** Decrypts, checks the body commitment, commits, returns a signed ack. blue ✓✓
- 4 Candidate submits** Seat shows a receipt code at once, even offline; the cell countersigns it later.
- 5 Shift closes** Entries go into a Merkle register; control signs the tree head; the witness co-signs it.
- 6 Anyone verifies** Candidate opens /verify offline: "Q17: record says C, the seat committed B" if anyone edited it.

Key problems with current CBT (JEE)

No layer-zero monitoring

Checks run in user space. A virtual machine or specialised relay software underneath the exam is invisible to it.

Web embed is vulnerable to backdoors

An embedded web client inherits the browser's attack surface: injected scripts, debug ports, patched pages.

Weak AES encryption

Symmetric keys shared along the chain protect secrecy only; nothing proves who wrote or who read a record.

What it costs today

- JEE Main 2025: a candidate's pop-up said 46 attempted, the response sheet said 29. It went to the Delhi High Court: their word against the vendor's PDF.
- CUET-UG 2026: a vendor-side glitch delayed a shift; 3,765 candidates needed a re-test.
- NEET-UG 2026: cancelled after a leak for ~22.7 lakh candidates; NEET moves to CBT from 2027.
- No candidate can check their own recorded answers.

Each problem, and how Saakshi answers it

Problem	Saakshi's answer	Status
No layer-zero monitoring (VM, relay software)	Kernel-level monitoring, like a game anti-cheat, under a pre-exam gate that already blocks named tools and scores VMs	Gate built; kernel next
Web embed open to backdoors	Native, fused seat app that refuses --inspect and remote-debugging; signed integrity policy; the network path is untrusted by design	Built
Weak AES encryption	End-to-end encryption plus signer + reader signatures on a git-like hash chain (GPG-style signed commits)	Built
System crashes, network drops	Relay-based store-and-forward: seat journal → centre relay → cell; resume on any seat; offline unlock code	Built
Server failure	Cluster of 3 independent data-centre cells; a wiped cell rebuilds from the relays with nothing lost	Built
Candidate cannot check the record	Receipt code at submit, offline /verify page, Merkle register co-signed by a witness	Built

What is new

Signer + reader signatures

Every answer is signed by the seat that wrote it and countersigned by the cell that read it, chained like git commits. Forgery or silent edits break the chain at a named entry.

Kernel-level exam monitoring

Anti-cheat techniques from online gaming applied to exams: see VMs, hooks and relay software below the app, not just process names.

Candidate-verifiable receipts

A UPI-style receipt code at submit and an 84 KB offline /verify page. No production CBT gives candidates this.

Decisions from evidence

A published rule re-tests only harmed candidates. Past history can corroborate a flag, never create one.

Current CBT vs Saakshi

	Current CBT (JEE)	Saakshi
Machine monitoring	User-space checks; VM and relay software can hide	Pre-exam gate now; kernel-level monitor next
Exam client	Web embed	Native fused app; debugging refused
Encryption	AES; keys shared along the chain	End-to-end to the cell; relay cannot read
Proof of each answer	None visible to the candidate	Seat signature + cell countersignature, hash-chained
Network outage	Centre server syncs later	Relay buffers; offline unlock code; outage predicted
Server failure	Single data-centre path	3-cell cluster, rebuild from relays, 0 lost
Disputed response sheet	Candidate's word vs vendor PDF	Receipt + offline /verify names the edited question
Re-exam decision	Ad hoc, often everyone	Published rule; only the harmed are re-tested

Who benefits, and how

Candidates

No acknowledged answer is ever lost. Each can verify their own record. No one is re-tested for someone else's failure; time lost is credited.

Governance (NTA, courts)

Court-ready evidence pack and audit trail, in time for the 5-month investigate-and-try window of the 2026 Amendment Bill.

Economic

Contained re-tests instead of nationwide ones save exam fees, travel and months of delay for lakhs of families.

Social & educational

Trust in exams restored. Look-alike groups (Hindi-medium, PwD, improvers) are protected; notices in English, Hindi and Tamil.

Measured impact on a 20,000-candidate exam

Re-tested 225 · Re-conducted 3 centres · Spared 18,994 · ₹ avoided 2,84,91,000

19,811

live candidates in the run

322

flags, each with a stated
reason

0.00%

honest candidates wrongly
flagged

813 / 1,563

took the blanket 2024 NEET
re-test offered

- Without evidence, every incident escalates to "re-test everyone"; in 2024 only about half of those offered a re-test took it.
- With Saakshi, a named human signs off a decision backed by per-candidate evidence.

Phased rollout

Phase 0 Today Working prototype: seat, relay, 3-cell cluster, control, witness. macOS + Windows CI.	Phase 1 Shadow mode Run beside the live system on one small NTA exam; compare records, measure real WAN.	Phase 2 Harden Kernel-level monitor, HSM-held keys, pinned TLS, WORM archive, independent witness.	Phase 3 Scale Relays at every centre, cell cluster at NIC data centres; NEET-size load test.	Phase 4 Mandate Conformance kit in the CBT vendor RFP: any vendor must pass the same tests.
---	--	---	---	--

Roles: NTA owns policy, keys and decisions · NIC runs control and cells · vendors run centres · witnesses sit outside the operational path.

Why it can be implemented

Runs on today's centre PCs: seat uses 2.3% median CPU and 166 MiB.

Standard cryptography: SHA-256, ECDSA P-256, ECDH, XChaCha20-Poly1305, RFC 9162 Merkle proofs.

Open: Apache-2.0 (GoI open-source policy); frozen protocol with golden test vectors.

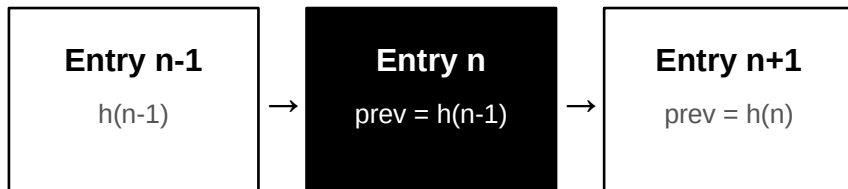
Fits policy direction: two-key paper release matches the Nilekani task force; NEET moves to CBT in 2027.

Risk	Mitigation
Kernel driver signing and stability	Microsoft/Apple signing programmes; staged rollout in shadow mode first
Control is a single point at T0 release	Replicate control; offline unlock code already works
Real WAN differs from lab	Phase 1 measures throughput and recovery on real links
Analytics tuned on synthetic data	Field pilot; humans decide every flag

Architecture and technology choices

Layer	Technology	Responsibility
Seat	Electron + React exam app, fused build	Signed journal (fsync), offline mode, integrity probes, on-device face presence; kernel monitor (next)
Centre relay	Bun server, MODE=relay, SQLite (WAL, full fsync)	Untrusted store-and-forward; invigilator console; spare relay takes over
Cell cluster ×3	Bun server, MODE=cell, sharded by centre	Decrypt, verify chain, group commit, countersign; REBUILDING from relays
Control	Bun server, MODE=control	Custody release, incidents, Merkle register (STH), audit, evidence pack
Witness	Bun server, MODE=witness	Co-signs only register heads with a valid consistency proof
Analytics	Python (numpy, scipy, networkx)	Cheating radar, decision engine; reads the cells' own export

E2E encryption with signer + reader hash



Like git: each entry commits to the one before it.

Signer: seat signs each header with its own key (ECDSA P-256), like a GPG-signed commit.

Reader: cell countersigns the entry hash it opened and verified: proof of who read it.

End-to-end: body sealed to the cell (ECDH + HKDF + XChaCha20-Poly1305); the relay carries ciphertext.

Register: Merkle tree head signed by control, co-signed by the witness.

Scale and redundancy

- Cells sharded by centre: each is its own failure domain.
- Any cell can be killed or wiped and rebuilt from relays: sent = verified = stored.
- Relay dies → spare relay; seats resync from their journals.
- Relay → cell ingest: p50 44 ms, p99 168 ms; 227,882 entries stored, 0 rejected at 20k candidates.
- Keys live outside databases; a wiped cell's old acks stay valid.

Five live demo acts on real processes

Act	What we do	Result
1 · Before	Seat runs AnyDesk and an overlay tool	Gate names both and blocks; green once closed
2 · T0	Cut Centre 42's link, release paper keys	2-of-3 custodians; phoned code unlocks only Centre 42; forged key rejected
3 · During	Kill and wipe a data-centre cell; force-quit a seat	Candidates keep answering; rebuilt with 0 lost; seat resumes elsewhere
4 · After	Insider edits an answer in the cell database	Audit locates it; /verify: "Q17: record says C, the seat committed B"
5 · Decide	Analytics on the cells' own export	Only harmed candidates re-tested; human sign-off verified

Measured numbers

Measure	Result	Evidence (docs/evidence/)
Ingest latency, relay to cell	p50 44 ms · p99 168 ms	stage7-load.txt
Stored / rejected at 20k candidates	227,882 entries · 0 rejected	stage7-load.txt
Candidates unlocked	19,803 / 19,803	stage7-load.txt
Chaos drill, 20 runs	0 of 16,000 answers lost · RTO medians 0.96–1.23 s	stage8-chaos.txt
Offline-start act	308 candidates, 7 centres, 6,586 entries	stage8-act2.txt
Tamper lab	1,000 / 1,000 byte flips located	stage0-tamper-lab.txt
Windows CI	Packaged seat end-to-end; renamed AnyDesk.exe blocked	stage5-gate-selftest.json

Five reasons

- | | | |
|----------|----------------------------------|---|
| 1 | Built, not drawn | A working system with tests, CI on Windows and five scripted live demos. Break anything and watch it recover. |
| 2 | Every claim is cited | Each number traces to an evidence file; limits are written down in a claims ledger and threat model. |
| 3 | Solves the real failures | JEE's response-sheet dispute, CUET's vendor glitch, NEET's blanket re-test: each maps to a built feature. |
| 4 | A protocol, not a product | Open, frozen spec with a conformance kit. NTA can hold any vendor to it. |
| 5 | Covers all 12 focus areas | Monitoring to AI risk analytics: 7 built, 5 partly built, each traced to code and tests. |

We can't promise zero failures. We make every failure recoverable, contained, provable.

Recoverable

No acknowledged answer is ever lost.

Contained

Only the harmed are re-tested.

Provable

Every candidate can verify their record.

Our ask: a shadow-mode pilot on one NTA exam, and the Saakshi conformance kit in the next CBT vendor RFP.